

# Jason (Jingxuan) Liu

Irvine, CA | jasonliu131060@outlook.com | (949) 617-9378 | [Portfolio](#)

## EDUCATION

**University of California, Los Angeles (UCLA)**

Expected Jun 2029

B.S. in Electrical Engineering | GPA: 3.97/4.00

**Completed Coursework:** Signals and Systems, Probability and Statistics, Foundations of Computer Vision

**Planned / In Progress:** Circuit Theory I & II Honors, Logic Design of Digital Systems, Analog Electronics I, Principles of Feedback Control, Digital Signal Processing, Introduction to Machine Learning

## PROFESSIONAL EXPERIENCE

**Undergraduate Researcher** | UCLA Integrated Sensors Laboratory

Jun 2026 – Aug 2026

Summer Undergraduate Research Program

- Built a dual-camera edge-AI wearable for LLM-based scene summarization, coupling a Himax WiseEye 2 for continuous low-res detection with an ESP32-S3 and high-res sensor for BLE image transfer to a companion app for Gemini API analysis
- Trained and deployed an INT8-quantized YOLOv5n model (1.7M parameters) to a Himax WiseEye 2 AI processor, achieving 0.833 mAP<sub>50</sub> across four Open Images V7 classes using PyTorch
- Calibrated cross-camera perspective via OpenCV homography, mapping detection bounding boxes to an OV5640 high-res sensor and tuning exposure, gain, and dynamic cropping to minimize BLE payload size
- Ported an ArduCAM OV2640 SPI burst driver (8 MHz) to Zephyr/nRF Connect SDK; measured 0.37 J/image over nRF52840 BLE versus 2.41 J/image over ESP32 Wi-Fi using a Power Profiler Kit II

## ENGINEERING EXPERIENCE

**Powertrain Project Engineer** | Bruin Racing – Supermileage Vehicle

Jun 2026 – Present

*Design and simulation phase; hardware build pending*

- Simulated a 120 W, 200 kHz active-clamp forward converter in LTspice to supply an isolated 12 V rail from the vehicle's 48 V bus; achieved  $< 94 \text{ mV}_{\text{pp}}$  output ripple ( $< 0.8\%$ ) across eight simulated load points spanning 0.25–10 A
- Modeled a PQ32/30 N97 transformer with 6:4 turns and  $L_m = 180 \mu\text{H}$  in SPICE; varied winding leakage and switching dead times (45–420 ns) to characterize hard switching and MOSFET voltage stress
- Designed mixed-signal sensing front ends for converter protection using an STM32G474, specifying Kelvin-sensed shunts, unity-gain-buffered dividers, and NTC circuits for on-chip ADC sensing and comparator-driven fault shutdown
- Simulated a two-switch forward-converter alternative that used input-rail clamping to limit switch voltage stress, targeting the Shell Eco-marathon 60 V maximum-voltage requirement

**Low-Voltage Subteam Member** | Bruin Racing – Supermileage Vehicle

Sep 2025 – Jun 2026

- Traced CAN messages with custom MCU diagnostics and repaired reversed connector pinouts and damaged wiring at the 2026 Shell Eco-marathon in Indianapolis, restoring communication across front, rear, and UI boards
- Designed an STM32F446RE rear-control board for torque, pressure, and acceleration sensing, rear-lighting control, and CAN communication with the low-voltage system of the Supermileage Vehicle using KiCad
- Evaluated Hall-effect, hydraulic-pressure, and strain-gauge brake-light sensing; selected and tested a pedal-mounted strain gauge

**Micromouse** | UCLA IEEE

Sep 2025 – May 2026

- Designed and assembled a 2-layer STM32F205 PCB integrating dual H-bridge motor control, magnetic encoder feedback, four-direction IR wall sensing, power regulation, and debug interfaces using KiCad
- Configured STM32 timers for quadrature encoder capture and four-channel motor PWM, implementing a 1 kHz closed-loop PD controller for straight-line motion and 90-degree turns
- Implemented time-multiplexed IR sensing with MOSFET-switched emitters, 12-bit ADC acquisition, and DMA averaging across 128 samples per reading to improve wall-detection stability
- Developed a  $16 \times 16$  flood-fill maze solver in C with wall maps and breadth-first distance updates; the completed Micromouse detected walls and traversed several maze blocks

## SKILLS

**Prototyping & Test Equipment:** PCB design, oscilloscope testing, soldering (hand, reflow and hot plate), simulation

**Microcontrollers & Embedded Platforms:** STM32, RP2040, ESP32, nRF52840, Arduino, Himax WiseEye 2

**Protocols & Controls:** I2C, SPI, BLE, PWM, PID Control, CAN

**Software & Tools:** STM32CubeIDE, ESP-IDF, nRF Connect SDK, OpenCV, KiCad, LTspice

**Programming:** C, C++, Python, MATLAB, R